

L17: Recap of Part II

Rules of probability

power calculation example

Practice problems

Rules of probability

Review of probability rules

Probabilities are numbers between 0 and 1.

$$0 \leq P(A) \leq 1$$

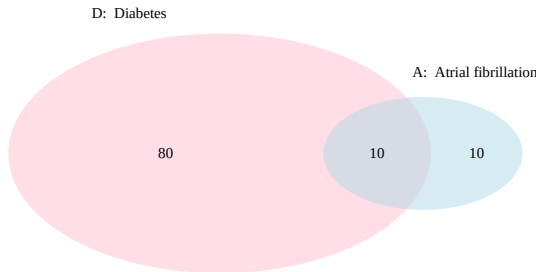
The probabilities in the sample space must sum to 1.

The probabilities of an event and its complement must sum to 1

$$P(A) + P(\bar{A}) = 1$$

Venn diagrams

If there are 180 total people in this study, what is the number missing from our sample space?



Adding and decomposing probability

For any two events A and B , $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

So what is the union of Atrial fibrillation and Diabetes in this example $P(A \cup D)$?

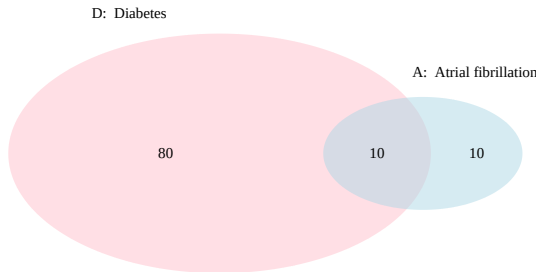
Adding and decomposing probability

For any two events A and B , $P(A) = P(A \cap B) + P(A \cap \bar{B})$

What would this look like in our example?

Venn diagram

There are 180 total people in this study. The intersection shown here is not included in the other pieces of the diagram.



Rules for independence

Written out in probability notation, for any two events A and B, the events are independent if:

$$P(A|B) = P(A)$$

or

$$P(B|A) = P(B)$$

or

$$P(A \cap B) = P(A) * P(B)$$

Rules for independence

Rules of probability

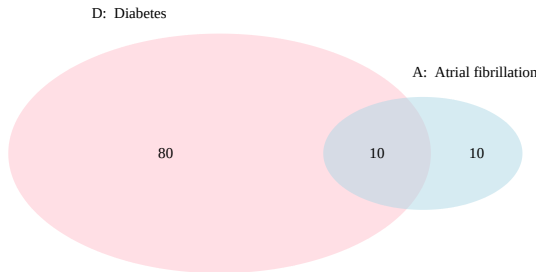
power calculation example

Practice problems

In our example is Atrial fibrillation independent of Diabetes?

Venn diagram

There are 180 total people in this study. The intersection shown here is not included in the other pieces of the diagram.



Multiplication rule and conditional probability

For any two events, the probability that both events occur is given by:

$$P(A \cap B) = P(B|A) \times P(A)$$

When $P(A) > 0$, the conditional probability of B, given A is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Bayes' theorem (simple version)

Suppose that A and A^c are disjoint events whose probabilities are not 0 and add exactly to 1. That is, any outcome has to be exactly in one of these events. Then if B is any other event whose probability is not 0 or 1,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

Example calculations

Let's say a group of 10 friends are deciding between Majors at Berkeley.

If I choose a science major, the probability of early morning lectures is .6

In this group 6 students choose science majors and 4 students choose language/literature related majors

I randomly choose one student to interview

The probability that the student I talk to has chosen a language/literature related major and has early morning lectures is 0.2.

What is the probability that students in language/literature related majors have early morning lectures?

Example calculations

From the information given I can know the following:

$$P(\text{Science major}) = .6 \quad P(\text{Language/literature major}) = .4$$

$$P(\text{Early lectures} \mid \text{Science major}) = .6$$

$$P(\text{Early lectures} \cap \text{Language/literature major}) = .2$$

Example calculations

Putting the known information into a table we have this:

Early lectures	Science Majors	Language/Literature majors	Totals
Yes	3.6	2	
No			
Totals	6	4	10

and we can figure out that of the 4 Language/Literature majors, if 2 have early lectures, then 2 must not have early lectures, so 50% of those who are Language/Literature majors have early lectures.

Example calculations - tree

The same information as a tree. Each branch splits on major first, then on early lectures.

Branch	Probability
Science → early	$0.6 \times 0.6 = 0.36$
Science → not early	$0.6 \times 0.4 = 0.24$
Language/Literature → early	$0.4 \times 0.5 = 0.20$
Language/Literature → not early	$0.4 \times 0.5 = 0.20$

The four branch probabilities sum to 1, which is the check that the tree is complete.

Conditional probabilities of Screening tests

Test result	Samples with known		Totals
	Disease	Samples without Disease	
Positive	90	8	98
Negative	14	96	110
Totals	104	104	208

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Practice problems

Two characteristics that are conditional on true disease status

- ▶ Sensitivity = $P(\text{Test positive} \mid \text{Disease})$
- ▶ Specificity = $P(\text{Test negative} \mid \text{No Disease})$

Two characteristics that are conditional on test result

- ▶ Predictive value positive = $P(\text{Disease} \mid \text{Test positive})$
- ▶ Predictive value negative = $P(\text{Not disease} \mid \text{Test negative})$

Conditional probabilities of Screening tests

What happens to sensitivity if we are in a context where the disease is more prevalent?

What happens to predictive value positive?

Distribution	Defined by:	Type of outcome	R notation
Normal	Mean and SD	Continuous	norm
Binomial	number and p	Binary (success or failure in n trials)	binom
Poisson	mean (λ)	Discrete count of events in an interval	pois

You should be familiar with what these distributions look like and what changes in the shape of the distribution as the key parameters change.

Which distribution?

What distribution would you think of for the following studies?

- ▶ Tracking the incidence of influenza during the weeks of winter
- ▶ Estimating the proportion of male and female children in a school who missed at least one day of school due to flu
- ▶ Estimating the number of minutes of exercise among students before and after New Year's day

Which distribution?

- ▶ Tracking the incidence of influenza during the weeks of winter → **Poisson**, a count of events in a fixed interval of time
- ▶ Estimating the proportion of male and female children in a school who missed at least one day of school due to flu → **Binomial**, a yes or no outcome over a fixed number of children
- ▶ Estimating the number of minutes of exercise among students before and after New Year's day → **Normal**, a continuous measurement

What proportion of adult women in the United States are taller than Beyonce?

In the US the mean height is 5'5" with a sd of 3.5"

Beyonce is 5'7" tall.

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Practice problems

#code to calculate - fill in during class

#_norm(____, _____, _____, option)

What is the Z- value for Beyonce's height?

$$Z = \frac{x - \mu}{sd}$$

How would we use the Z value to calculate the probability of being **shorter** than Beyonce?

```
#code to calculate taller than Beyonce using measured height  
pnorm(q=67, mean=65, sd=3.5, lower.tail=F)
```

```
## [1] 0.2838546
```

```
# code to calculate shorter using Z value  
# to do during class
```

calculations with the normal distribution

How many women are taller than Ariana Grande (5' 0") and shorter than Beyonce?

```
#code to calculate taller than Beyonce using measured height  
pnorm(q=67, mean=65, sd=3.5, lower.tail=F)
```

```
## [1] 0.2838546
```

```
# code to calculate proportion in range  
# to do during class
```

Imagine you are working at an aquarium shop. You have a tank with 600 guppies, 30% of which have black spots on their tail.

You have a client who wants to take home 4 guppies, 2 with black spots and 2 without black spots.

You can net 4 fish at a time. What is the probability of netting the fish your client wants in any attempt?

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Practice problems

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Normal approximation for binomial distributions

Suppose that a count X has the binomial distribution with n observations and success probability p . When n is large, the distribution of X is approximately Normal. That is,

$$X \sim N(\mu = np, \sigma = \sqrt{np(1-p)})$$

As a general rule, we will use the Normal approximation when n is so large that $np \geq 10$ and $n(1-p) \geq 10$.

It is most accurate for p close to 0.5, and least accurate for p closer to 0 or 1.

If X has the Poisson distribution with mean number of occurrences per interval μ , the possible values of X are 0, 1, 2, If k is any one of these values, then

$$P(X = k) = \frac{e^{-\mu} \mu^k}{k!}$$

The rate of measles in California is roughly 1.75 cases per month, usually from travelers exposed while outside of the country. Between December 2014 and April 2015 the rate was roughly 26.2 cases per month.

What is the probability of observing exactly 2 cases in a normal month? (worked out by hand, then confirm with r)

##fill in during class

What is the probability of observing 0, 1 or 2 cases in a normal month? (there is more than one way to do this)

Calculations with Poisson

```
dpois(0,1.75) + dpois(1,1.75) + dpois(2,1.75)
```

```
## [1] 0.7439697
```

```
ppois(2,1.75)
```

```
## [1] 0.7439697
```

What is the probability of observing 26 cases or more in a normal month? Would you feel comfortable calling this an outbreak?

fill in during class

“26 or more” is the upper tail, so we want $P(X \geq 26) = 1 - P(X \leq 25)$.

```
# lower.tail = FALSE keeps the precision in the far tail  
ppois(25, 1.75, lower.tail = FALSE)
```

```
## [1] 9.602482e-22
```

Writing it as `1 - ppois(25, 1.75)` returns exactly 0, because the probability is far smaller than R can represent as a difference from 1. The answer is not 0, it is about 10^{-21} . Under the usual rate, 26 cases in a month is essentially impossible, so yes, we would call this an outbreak.

μ and p are population parameters for the mean and proportion. There is one unique value for μ and p in the underlying population.

$\bar{x}$ and $\hat{p}$ are statistics computed using samples. We refer to them as the sample mean and sample proportion, respectively. If we change the sample our statistics will likely also change. Statistics vary across samples.

Sampling distribution of a sample mean for a Normal population

- ▶ If individual observations have a $N(\mu, \sigma)$ distribution, then the sample mean $\bar{x}$ of a simple random sample of size n has a $N(\mu, \frac{\sigma}{\sqrt{n}})$

You should be able to think through what happens when we adjust parts of this equation. (ie what happens to variability of the sample mean when we increase n ?)

We can use the Central Limit Theorem to treat the distribution of a sample mean as normally distributed under conditions when the underlying population values are not normally distributed.

The Central Limit Theorem (CLT)

Draw a simple random sample of size n from any population with mean μ and finite standard deviation σ . When n is large, the sampling distribution of the sample mean $\bar{x}$ is approximately Normal:

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The CLT allows us to use Normal probability calculations to answer questions about sample means from many observations (questions relying on the sampling distribution of the sample mean) even when the population distribution is not Normal.

Sampling distribution of the proportion $\hat{p}$

- ▶ The mean of the sampling distribution is p , the population parameter
- ▶ The standard deviation of the sampling distribution is $\sqrt{\frac{p(1-p)}{n}}$
- ▶ As the sample size increases, the sampling distribution of $\hat{p}$ becomes approximately Normal. This is the Central Limit Theorem for a proportion!
- ▶ For this to apply, we require:
 - ▶ the population is at least 20 times as large as the sample
 - ▶ both np and $n(1-p)$ are larger than 10.

We can use information about the variability of sample means to generate confidence intervals and p values for our estimates and begin to use this information to draw inference from our data.

Confidence intervals for the mean μ

Confidence level C	90%	95%	99%
Critical value z^*	1.645	1.960 (≈ 2)	2.576

- These numbers correspond to the value on the x-axis corresponding to having 90%, 95%, or 99% of the area under the Normal density between $-z$ and z .

The generic format of a confidence interval is then:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

You should know how to create a confidence interval and what changes to your study/data would cause the confidence interval to be larger or smaller.

Define the Hypothesis

A **Null Hypothesis (H_0)** is the hypothesis that is assumed to be true and the start of a test. This is often expressed as a statement of equality (ie. mean equal to a certain value or no difference between groups)

An **Alternative Hypothesis (H_A)** is usually the inverse of the null hypothesis and is expressed as a statement of difference.

- ▶ H_A : The mean is greater than the Null (one tailed)
- ▶ H_A : The mean is less than the null (one tailed)
- ▶ H_A : The mean is not equal to (greater or less than) the null (two tailed)

When we test a hypothesis, we are not trying to prove H_A , we are trying to **disprove H_0**

Decide on a threshold for rejecting the null

We choose a probability that we decide is small enough that we are unlikely to have observed it by chance if H_0 is true.

This threshold is our α .

We must decide if our hypothesis is one-tailed or two-tailed

You should be able to read a description of a study or hypothesis test and know whether they hypothesis test would be one tailed/one sided or two tailed/two sided.

P-value: The probability, assuming that H_0 is true, that the test statistic would take a value at least as extreme (in the direction of H_a) as that actually observed. The smaller the p-value, the stronger the evidence against H_0 provided by the data.

Type I error, and Type II error in hypothesis tests

You should know the difference between type I and type II error and what would cause error or power to increase or decrease

	H_a is true	H_0 is true
Reject H_0	Correct decision	Type I error (α)
Fail to reject H_0	Type II error (β)	Correct decision

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Example

Here we will go through another example from the Baldi and Moore textbook. This example assumes you are planning a quality control study to look at whether storage impacts the perceived sweetness of a beverage. Ten professional tasters will rate the sweetness on a 10 point scale before and after storage. We know that the standard deviation of sweetness ratings is $= 1$. We also know that a mean sweetness change of 0.8 units on this scale is noticed by consumers. We want 90% power and an alpha of 0.05 for our study. We have a set of 10 values representing the difference in sweetness caused by storage. Let's presume that we only care about a perceptible *loss* in sweetness.

What is the null hypothesis here?

What is our alternative?

Is our hypothesis one or two sided?

Example

What is the null hypothesis here? no difference in sweetness or $\mu=0$

What is our alternative? μ decreases by .8 or more

Is our hypothesis one or two sided? one sided (we only care about decreases)

Example

We will start by finding the Z alpha:

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

```
qnorm(.05)
```

```
## [1] -1.644854
```

$$-1.645 = \frac{\bar{x} - 0}{\frac{1}{\sqrt{10}}}$$

Solve this for $\bar{x}$

$$\bar{x} = -1.645 \times \frac{1}{\sqrt{10}} = -0.520$$

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Alternate calculation

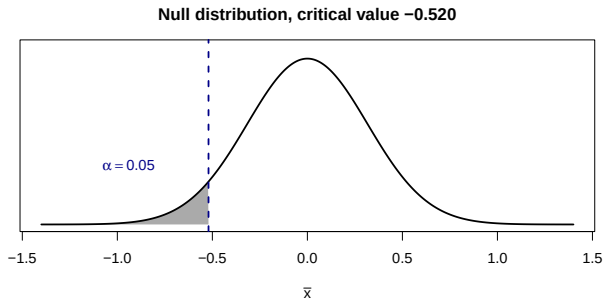
Note that you could also calculate the cutpoint value in units of $\bar{x}$ directly in R by setting the mean and sd

```
qnorm(.05, mean=0, sd=(1/(sqrt(10))))
```

```
## [1] -0.5201484
```

null distribution

So here we have our null distribution with the value at which we reject the null



The shaded area is 5% of the null distribution. What is our β ?

Example

We must choose a value at which to evaluate power. Here we will choose an alternate hypothesis that the mean sweetness difference is -0.8 . Since we know a sample mean greater than -0.520 causes us to fail to reject H_0 we need to calculate the proportion of a distribution centered at -0.8 that would be below this value.

Example:

Using R to calculate the probability, relative to a sampling distribution centered at the alternative value -0.8

```
pnorm(-0.520, -.8, (1/sqrt(10)))
```

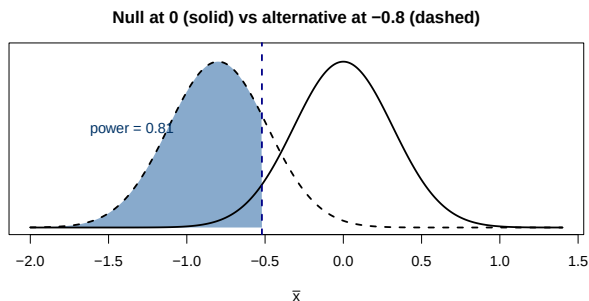
```
## [1] 0.8120397
```

Thus power or $P(\text{reject null}(0) | \text{Null is false (true sweetness change is } -0.8))$ is ~ 0.81

Remember that Power is $1 - \beta = P(\text{reject null} \mid \text{null is false})$

In this example, β is $1 - 0.81$ or ~ 0.19

Example: the same picture



Power is the shaded part of the **alternative** curve, not the null curve.

To think about sample size for a z-test (or more generally for any test), four things matter:

- ▶ **Significance level α** : How much protection do we want against getting a statistical significant results from our sample when there really is no effect in the population?
- ▶ **Effect size**: How large an effect in the population is important in practice?
- ▶ **Power ($1 - \beta$)**: How confident do we want to be that our study will detect an effect of the size we think is important? I.e., what is the probability of rejecting H_0 when the alternative hypothesis is true?
- ▶ **variability in the population**: Remember that the underlying variability in our population affects the variability of our sample mean

When we are calculating sample size the steps we follow are:

- ▶ Find the Z_α and use this to calculate the value of our variable at which we would reject the null.
- ▶ Find Z_β and use that to calculate what value this would be on the curve of the alternative hypothesis.
- ▶ Set these two expressions equal to each other and solve for n .

Practice problems

How to use this section

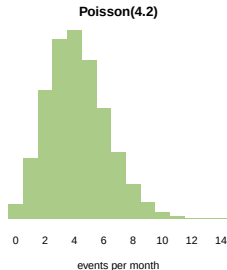
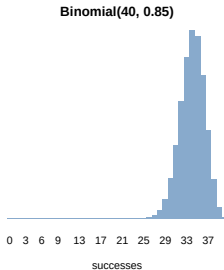
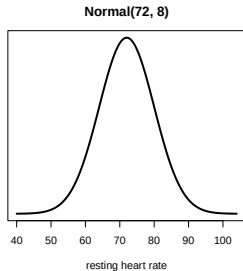
Seven problems. They are deliberately **not** one problem per topic. Most of them make you combine two or three ideas, and several ask you to do something other than compute, because on a real analysis the hard part is usually deciding what to compute and whether the output means what you think it means.

The tasks you will meet here.

- ▶ Find the errors in somebody else's work
- ▶ Carry one messy scenario through several different tools
- ▶ Read R output and decide which claims it supports
- ▶ Start from the answer and work back to the input
- ▶ Rank study designs without computing anything
- ▶ Explain a result in plain language to a colleague

Attempt each one before the answer slide. A worked solution you read feels like learning and mostly is not.

Shapes to recognize



Know which parameter moves the centre and which one moves the spread, and know which way each distribution leans.

A. Find the errors

A clinic records a mean of 3.1 medication errors per week. A student was asked to (i) find the probability of **at least 5** errors in a week, and (ii) give a 95% CI for the mean weekly count from 40 weeks of records, with $\bar{x} = 3.1$ and known $\sigma = 1.8$.

Here is their submitted work. **There are three errors. Find them.**

```
# (i) at least 5 errors in a week
```

```
1 - ppois(5, lambda = 3.1)
```

```
## [1] 0.0943
```

```
# (ii) 95% CI for the mean
```

```
3.1 + c(-1, 1) * 1.96 * 1.8
```

```
## [1] -0.428 6.628
```

```
# (iii) conclusion
```

```
# "There is a 95% probability that mu lies between -0.43 and 6.63."
```

A. Answer, the two calculation errors

Error 1, the boundary. “At least 5” is $X \geq 5$, whose complement is $X \leq 4$, not $X \leq 5$. The student computed $P(X \geq 6)$.

```
1 - ppois(4, lambda = 3.1)      # correct
```

```
## [1] 0.2018105
```

```
1 - ppois(5, lambda = 3.1)      # what was submitted
```

```
## [1] 0.09433383
```

The answer is 0.20, not 0.09. Being one off more than doubled it.

Error 2, the standard error. The interval must use the standard deviation of $\bar{x}$, which is $\sigma/\sqrt{n}$, not σ .

```
3.1 + c(-1, 1) * 1.96 * 1.8 / sqrt(40)
```

```
## [1] 2.542174 3.657826
```

So 2.54 to 3.66, not -0.43 to 6.63 .

A. Answer, the interpretation error

Error 3. “There is a 95% probability that μ lies between 2.54 and 3.66” is the wrong reading, even with the corrected numbers.

μ is a fixed number. It is not random, so it does not have a probability of being anywhere. What is random is the interval, because it is built from a sample that could have come out differently.

The defensible statement is about the **procedure**. If we repeated this study many times and built an interval each time the same way, about 95% of those intervals would contain the true μ . This one either does or does not, and we do not get to know which.

Two more claims that would also have been wrong here, because they show up constantly.

- ▶ “The CI shows a 95% chance the null is true.” A CI says nothing about the probability of a hypothesis.
- ▶ “ μ is probably near the middle of the interval.” Every value in the interval is equally compatible with the data at this confidence level.

B. One program, three tools

A district TB contact-tracing program.

Stage 1. Among screened contacts, the prevalence of active TB is 0.06. The rapid test has sensitivity 0.80 and specificity 0.97.

- ▶ B1. Of everyone who tests positive, what fraction actually has TB?
- ▶ B2. Among 5,000 contacts screened, how many false positives are expected?

Stage 2. Everyone who screens positive is referred onward. On a clinic day, referrals arrive at a mean rate of 2.4 per hour.

- ▶ B3. What distribution fits the number of referrals in an hour, and why?
What is the probability of an hour with none?

Stage 3. The program tracks days from referral to treatment start. For 64 patients, $\bar{x} = 21$ days, and $\sigma = 6$ days is known.

- ▶ B4. Give a 95% CI for the mean time to treatment.

B. Answer, stages 1 and 2

B1. This is a predictive value, so the conditioning runs from test result back to disease.

$$(0.80 * 0.06) / (0.80 * 0.06 + 0.03 * 0.94)$$

```
## [1] 0.6299213
```

About 0.63. Notice this is far better than the 0.20 we would get at a 2% prevalence, and the reason is that contact tracing has already concentrated the risk. Who you choose to test matters more than small improvements to the test.

B2. False positives come out of the healthy group only.

$$5000 * (1 - 0.06) * (1 - 0.97) \quad \# \text{ false positives}$$

```
## [1] 141
```

$$5000 * 0.06 * 0.80 \quad \# \text{ true positives for comparison}$$

```
## [1] 240
```

B. Answer, stages 2 and 3

B3. Poisson. We are counting events in a fixed interval of time with no fixed number of trials, and $\lambda = 2.4$ per hour.

```
dpois(0, lambda = 2.4)
```

```
## [1] 0.09071795
```

About 0.09, so roughly one hour in eleven sees no referral.

B4. The outcome is now a continuous measurement, so we want an interval.

```
21 + c(-1, 1) * qnorm(0.975) * 6 / sqrt(64)
```

```
## [1] 19.53003 22.46997
```

19.5 to 22.5 days. One program, three different tools, and nothing labelled which was which. Deciding that is the skill.

C. What does this output support?

```
pbinom(3, size = 20, prob = 0.3)
## [1] 0.1071
dbinom(3, size = 20, prob = 0.3)
## [1] 0.0716
pbinom(3, size = 20, prob = 0.3, lower.tail = FALSE)
## [1] 0.8929
```

Which of these claims does the output support? (Select all that apply)

- a. The probability of exactly 3 successes is 0.0716.
- b. The probability of at most 3 successes is 0.1071.
- c. The probability of at least 3 successes is 0.8929.
- d. 0.1071 and 0.8929 are complements of each other.
- e. A Normal approximation to this binomial would be reasonable.

C. Answer

a, b, d and e. Only c is unsupported.

a is right. `dbinom` is the probability of exactly that value.

b is right. `pbinom` at 3 is everything at or below 3.

c is the trap. `lower.tail = FALSE` gives everything **strictly above** 3, so 0.8929 is $P(X \geq 4)$. To get $P(X \geq 3)$ you would need $1 - \text{pbinom}(2, \dots)$, which is 0.9645. Different question, different number.

d is right, and it is why c is wrong. At most 3 and more than 3 partition everything, so 0.1071 plus 0.8929 is 1.

e is right. Here $np = 6$ and $n(1 - p) = 14$, both at least 10.

The habit worth building. When you read R output, translate each line back into a sentence about the world before you use it. Most output errors are not arithmetic, they are somebody solving a slightly different problem than the one asked.

D. Work backwards

Both parts give you the answer and ask for the input.

D1. A 95% confidence interval for a mean needs to have a total width of 5.0, so ± 2.5 . The population standard deviation is $\sigma = 12$. How many observations do you need?

D2. A screening test has sensitivity 0.90 and will be used in a population with prevalence 0.05. What specificity would the test need for its positive predictive value to reach 0.50?

D. Answer

D1. Set the margin of error equal to what we want and solve for n .

$$2.5 = 1.96 \times \frac{12}{\sqrt{n}} \Rightarrow \sqrt{n} = \frac{1.96 \times 12}{2.5}$$

```
(1.96 * 12 / 2.5)^2
```

```
## [1] 88.51046
```

That gives 88.5, and sample size always rounds **up**, so $n = 89$. Rounding down would leave the interval slightly too wide.

D. Answer, part 2

D2. PPV of 0.50 means true positives exactly equal false positives.

$$0.90 \times 0.05 = (1 - \text{spec}) \times 0.95$$

$$1 - (0.90 * 0.05) / 0.95$$

```
## [1] 0.9526316
```

You would need specificity of about 0.953.

Sit with that. Even at 95.3% specificity, a coin flip is all a positive result is worth at this prevalence. Rare conditions demand extraordinary specificity, and this is the same lesson as the earlier screening slides, arrived at from the opposite direction.

E. Rank these four studies

Four planned studies, all two-sided tests of a mean with the same outcome variable and the same population standard deviation.

Study	n	α	Effect size to detect
1	50	0.05	0.4 SD
2	50	0.01	0.4 SD
3	200	0.05	0.4 SD
4	50	0.05	0.8 SD

E1. Rank them from lowest power to highest. Do not compute anything, reason from the picture on slide 54.

E2. Study 2 tightened α to 0.01. In plain terms, what did that buy, and what did it cost?

E. Answer

Study 2, then Study 1, then Studies 3 and 4 tied at the top.

Study 2 versus Study 1. Same everything except a stricter α . Tightening α slides the cutoff further out, which shrinks the shaded region under the alternative curve. So Study 2 has the least power.

Studies 3 and 4 versus Study 1. Study 3 quadruples n , which narrows both curves so they overlap less. Study 4 doubles the effect, which pushes the curves apart. Both raise power.

Now the part worth remembering. Studies 3 and 4 have **exactly** the same power. Power depends on the effect size multiplied by $\sqrt{n}$, and

```
0.4 * sqrt(200)
```

```
## [1] 5.656854
```

```
0.8 * sqrt(50)
```

```
## [1] 5.656854
```

E. Answer, part 2

E2. Study 2 bought a lower chance of a **Type I error**, claiming an effect that is not real. It paid for that with a higher chance of a **Type II error**, missing an effect that is real.

That is the trade, and it is forced. With n and the effect size fixed, α and β move in opposite directions. You cannot shrink both by choosing a different threshold, because the threshold is a single line on the picture and sliding it out of one curve's tail slides it into the other's.

The only lever that improves both at once is more data, which narrows both curves so they overlap less. That is what Study 3 did.

So when is a strict α worth it? When a false positive is the more expensive mistake. Approving a drug that does not work, or launching a costly program on a result that will not replicate. When the more expensive mistake is missing a real effect, for instance failing to detect a genuine environmental hazard, an α of 0.01 is buying the wrong protection.

F. Six scenarios

Name the distribution for each, and give its parameters.

1. Of 250 patients on a ward, the number who acquire an infection, each independently at risk 0.04.
2. The number of 911 calls in a 15 minute window, when calls average 7 per hour.
3. The birth weight of one randomly chosen newborn.
4. The number of mosquito bites a volunteer receives during a 30 minute exposure trial.
5. Of 12 water samples, the number exceeding a lead threshold.
6. The mean systolic blood pressure of a random sample of 100 adults.

F. Answer

1. **Binomial(250, 0.04)**. Fixed roster, independent yes or no, count the yeses.
2. **Poisson(1.75)**. Watch it. The rate is 7 per hour but the window is 15 minutes, so $\lambda = 7/4 = 1.75$. **Always rescale the rate to match the interval in the question.**
3. **Normal**. A continuous measurement, with parameters μ and σ from the population.
4. **Poisson**. Counting events in a fixed interval with no fixed number of trials.
5. **Binomial(12, p)**. Fixed number of samples, each over or under the threshold.
6. **Normal**, but for a different reason. This is a *sampling distribution*. By the CLT the mean of 100 observations is approximately Normal centred at μ with standard deviation $\sigma/10$, whatever the shape of individual blood pressures.

Two discriminators hidden in this list. Item 2 catches anyone who used the rate without rescaling. Item 6 catches anyone who did not notice that the quantity is an average rather than one measurement, which changes the spread by a factor of 10.

G. Answer a colleague

A colleague reads that a new test is “97% accurate” and concludes that if she tests positive, there is a 97% chance she has the condition. The condition affects about 1 person in 500.

Write two or three sentences back to her. Support your answer with a number.

G. Answer

Take “97% accurate” to mean sensitivity and specificity are both 0.97, and prevalence is $1/500 = 0.002$.

$$(0.97 * 0.002) / (0.97 * 0.002 + 0.03 * 0.998)$$

```
## [1] 0.0608532
```

A reply that would earn full marks.

The 97% describes how the test behaves when you already know whether someone has the condition. It does not describe what a positive result means. Because only 1 person in 500 has this condition, the 3% of false positives is drawn from a group about 500 times larger than the group of true cases, so those false positives swamp the real ones. Your actual chance of having the condition given a positive result is about 6%, not 97%.

The lesson. Reversing a conditional probability changes its value, often enormously, and how much depends on how rare the condition is.

Where to review

If a problem gave you trouble, this is where to go back to.

Problem	If you struggled, revisit
A	Slides 33 and 35 on cumulative boundaries, 37 on the standard error, 41 on what a CI means
B	Slides 12 and 17 on Bayes and screening, 19 on choosing a distribution, 41 on intervals
C	Slides 28 and 33 on what the d and p prefixes do, 29 on the approximation rule
D	Slides 37 and 41 on how $\sqrt{n}$ enters, 18 on prevalence and PPV
E	Slides 45 on the two errors, 51 through 55 on the power picture and its levers
F	Slide 19 on the decision rule, 38 on the CLT
G	Slides 17 and 18 on the direction of the conditioning bar

Four things to carry into the exam. The conditioning event sets the denominator. Sensitivity and specificity belong to the test, predictive values belong to the population. For at least k , subtract the cumulative at $k - 1$. Precision scales with $\sqrt{n}$, so four times the data buys half the width.